

Broadway Show analysis

Avril

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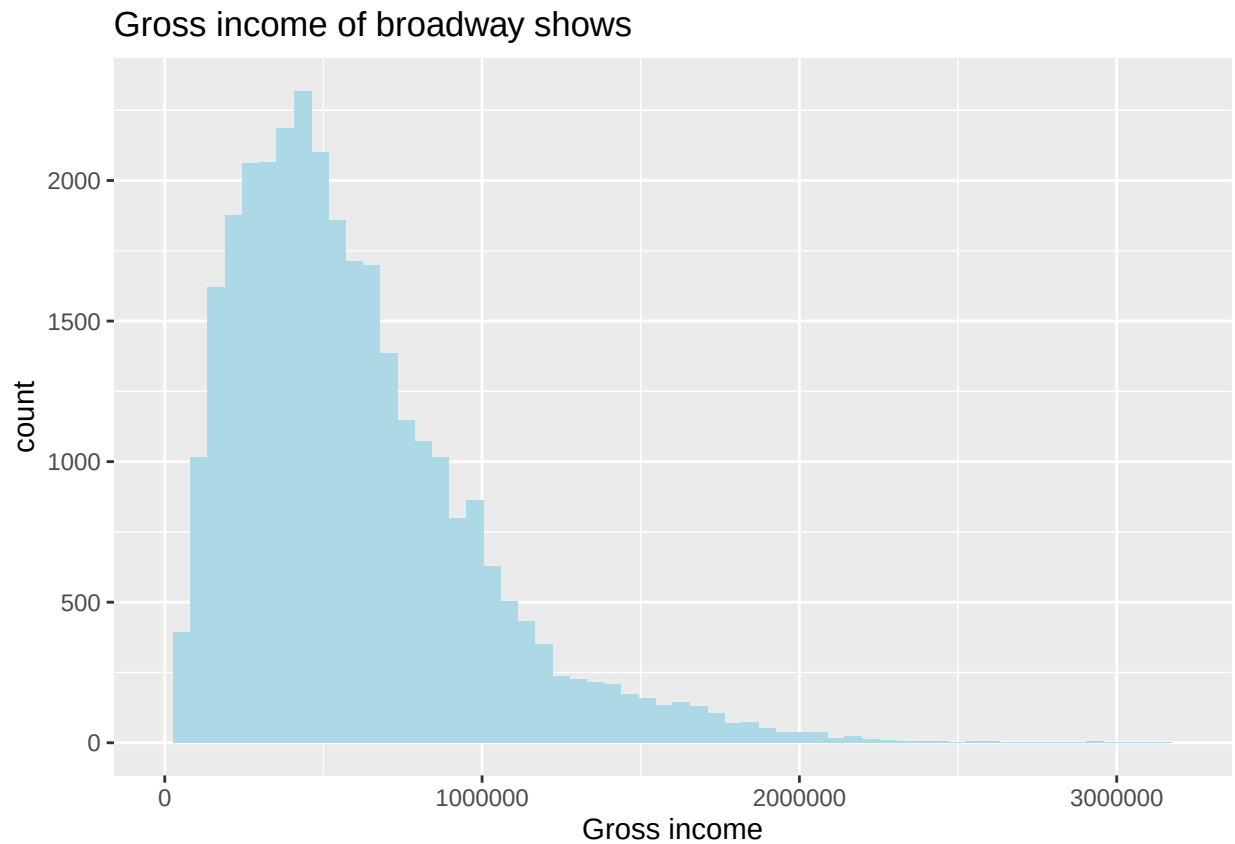
Introduction

This project analyzes a database of all Broadway shows from 1990 to 2016, which was collected from the official Broadway website. The focus/outcome variable will be on each show's **weekly gross revenue** (in dollars). The analysis examines whether show type, theater, and total attendance influence gross revenue. These predictor variables were selected due to their potential relationship with the choice variable.



Gershwin Theatre | Theater in Midtown West, New York. (n.d.). Time Out. <https://media.timeout.com/images/103258724/750/422/image.jpg>

Univariate distribution



This graph shows the gross revenue of Broadway shows. The mean revenue was \$ 596,069, with a standard deviation of \$ 380,980.

The highest grossing show was Wicked in the year 2013, generating a revenue of \$ 3,201,333.

The lowest grossing show was Marc Salem'S Mind Games in the year 2004, generating a revenue of \$ 1,037.

Table 1: Standard deviation of the revenue of Broadway shows by play type

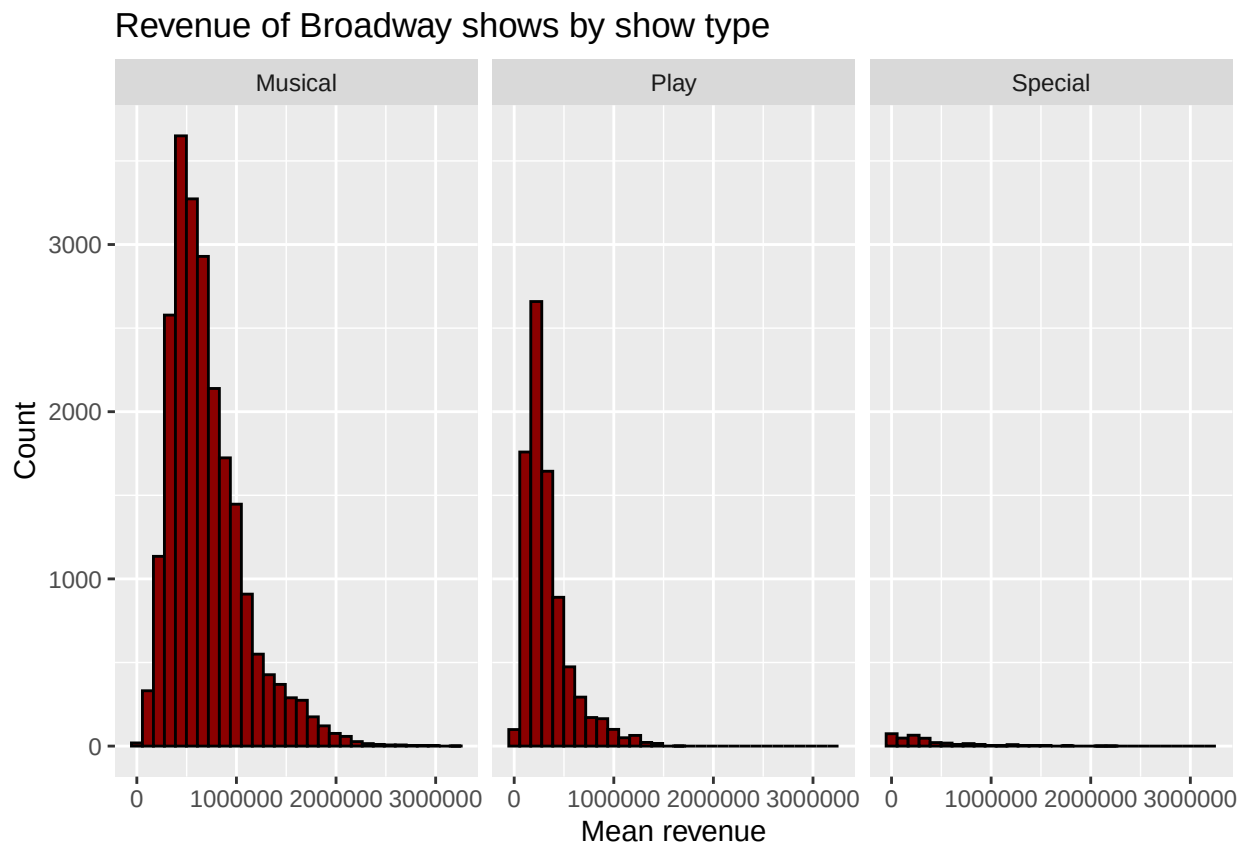
Show type	Standard deviation of Revenue
Musical	375,445
Play	226,715
Special	400,421

Table 2: Mean revenue of Broadway shows by show type

Show Type	Mean Revenue
Musical	700,312
Play	325,718
Special	365,322

Bivariate analyses

By show type



This graph shows the gross revenue of Broadway shows by show type. The tables show the mean revenue and the standard deviation of Broadway shows by show type. On average, musicals did better than plays and specials, with a mean revenue of \$ 700,312.

Table 3: Standard deviation of the revenue of Broadway shows by theatre location

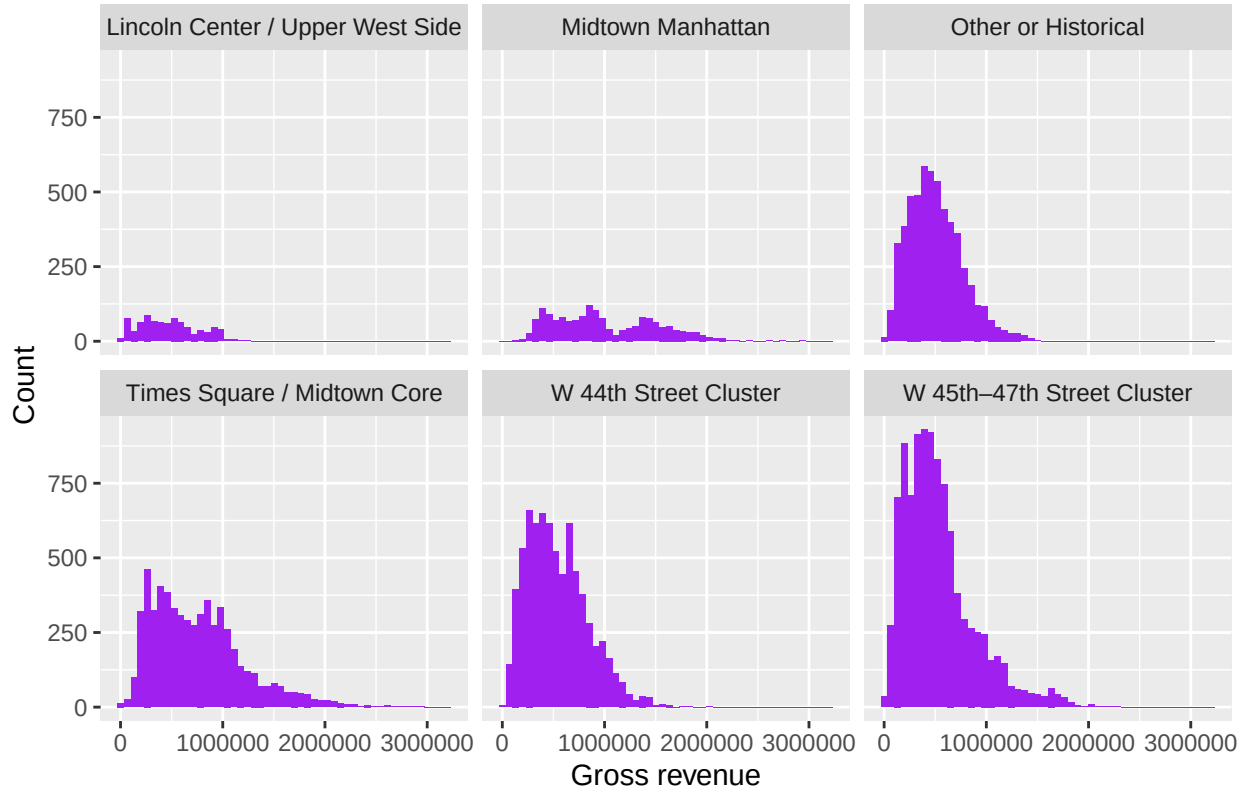
Theatre cluster location	Standard deviation of Revenue
Lincoln Center / Upper West Side	283,937
Midtown Manhattan	515,316
Other or Historical	264,682
Times Square / Midtown Core	452,987
W 44th Street Cluster	294,185
W 45th–47th Street Cluster	346,855

Table 4: Mean revenue of Broadway shows by theatre location

Theatre cluster location	Mean Revenue
Lincoln Center / Upper West Side	474,672
Midtown Manhattan	1,017,944
Other or Historical	505,416
Times Square / Midtown Core	758,347
W 44th Street Cluster	536,457
W 45th–47th Street Cluster	531,475

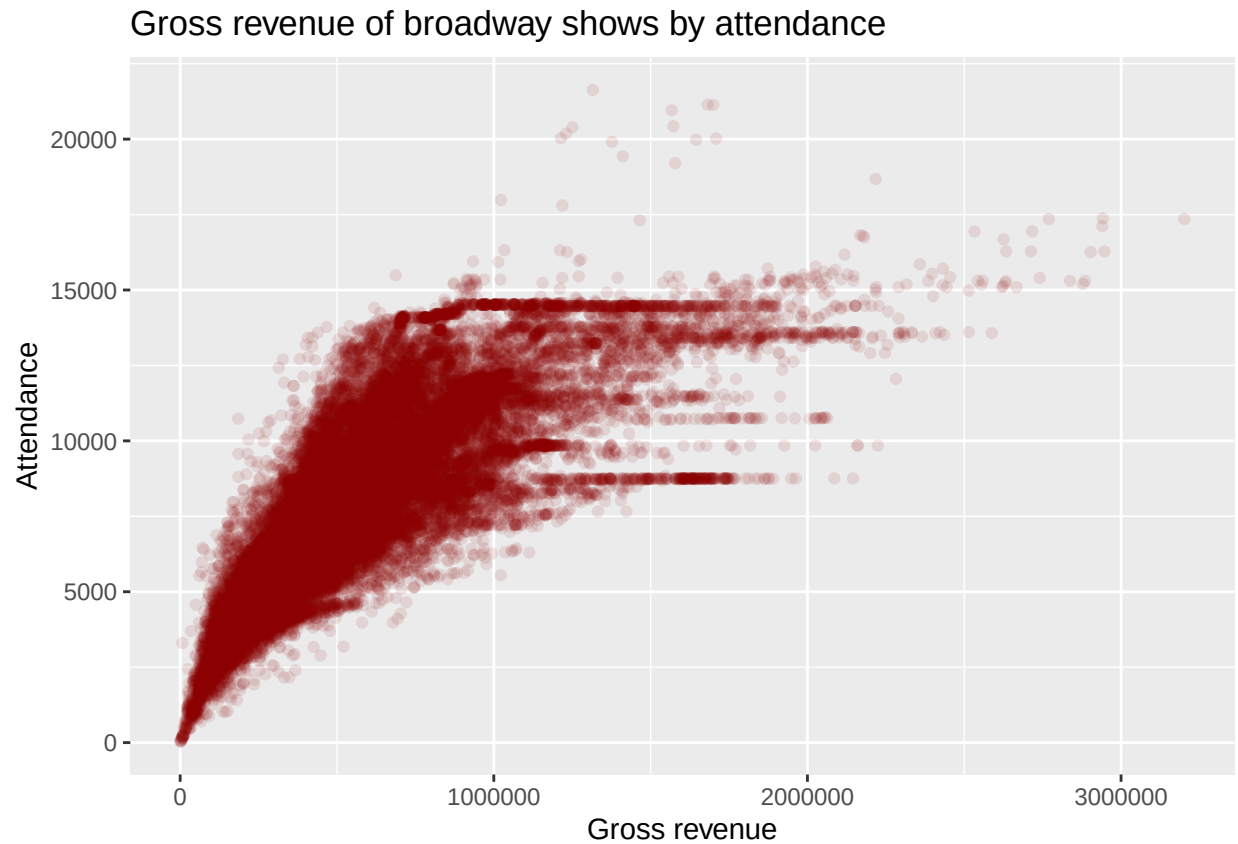
By theatre location

Broadway show gross revenue by theatre location



This graph shows the gross revenue of Broadway shows by theatre location. The tables show the standard deviation and mean revenue. On average, the Gershwin and Winter Garden theaters in Midtown Manhattan performed better in terms of gross revenue.

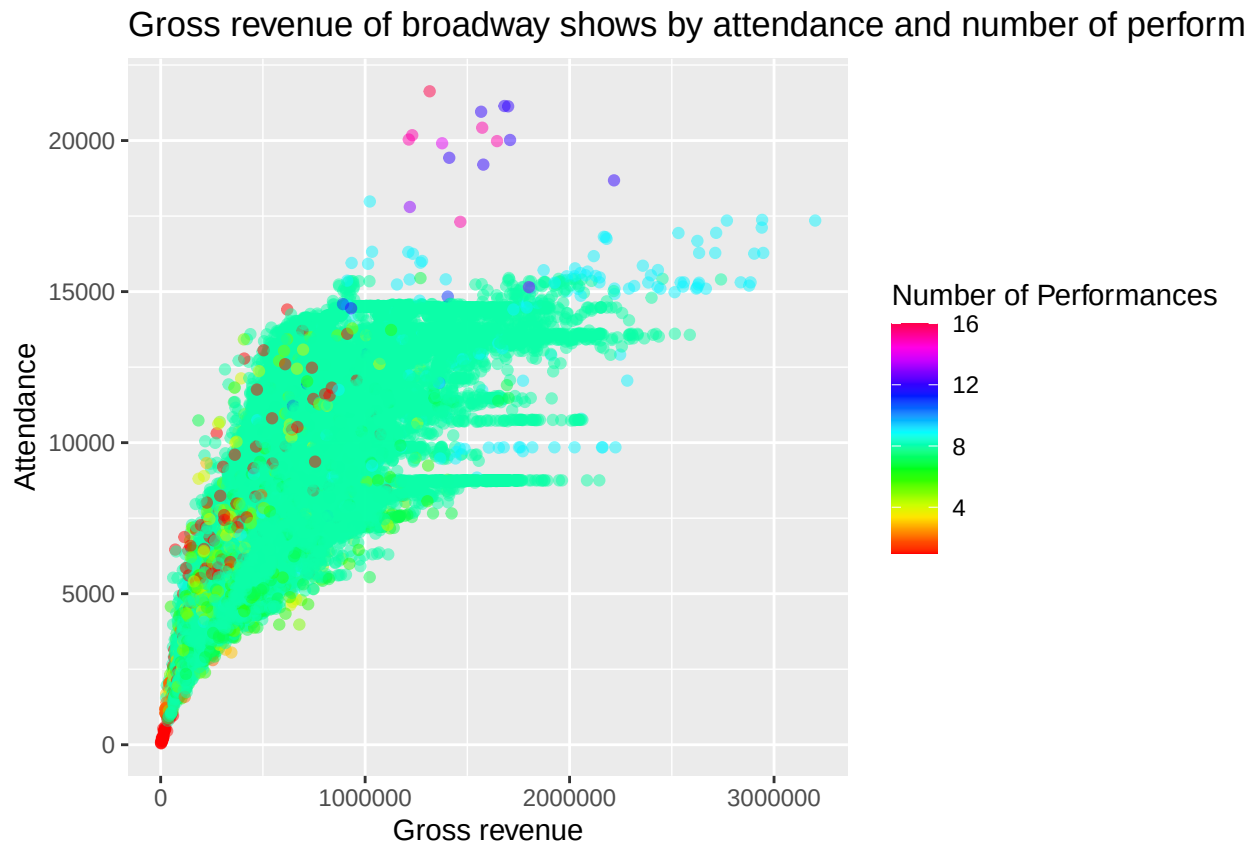
By attendance



This scatter plot shows the gross revenue of Broadway shows by total attendance. There is a strong correlation (0.79) between total attendance and gross revenue.

Multivariate analysis

By attendance and number of performances



The following scatterplot shows the gross revenue of Broadway shows by total attendance, but also taking into account the number of weekly performances. The showings with 0 performances were removed from the dataset for this plot; assuming that there is likely an error in the data collection for these showings. In general, most shows had between 7 and 9 showings, with 8 being the most common.

Conclusion

Musicals generally performed the best overall. Theaters in Times Square or Midtown Manhattan generated the highest revenue, and most shows had 7 or 8 performances per week. Wicked was the top-grossing show, and Marc Salem's Mind Games was the lowest grossing one. For future research, it would be valuable to more accurately track performance counts, as some shows frequently reported zero performances. Additionally, monitoring the average ticket price per attendee would provide deeper insights into revenue trends.

References

CORGIS Datasets Project. (n.d.). <https://corgis-edu.github.io/corgis/csv/broadway/>

Gershwin Theatre | Theater in Midtown West, New York. (n.d.). Time Out. <https://media.timeout.com/images/103258724/750/422/image.jpg>

The Broadway League | the official website of the Broadway industry. (n.d.). <https://www.broadwayleague.com/home/>